

Modeling of water-PCM solar thermal storage system for domestic hot water application using Artificial neural networks

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ARTICLE INFO

Keywords:

Artificial neural networks
Thermal storage
Solar domestic water heating
Phase change materials
Numerical modeling

ABSTRACT

Numerical modeling of solar thermal storage systems is often challenged with limitations on the computational effort due to their transient non-linear behavior that dictates accurate modeling of the physics over long-term operations (i.e., annual). Such challenges result into scarcity of literature on comprehensive design guidelines for solar thermal storage systems. This study presents a framework through which the potential of artificial neural network (ANN) modeling of a hybrid solar thermal storage system involving phase change materials is extensively investigated. An experimentally validated numerical model for the system is used to generate the training and testing datasets for the ANN model. The effect of changing the sampling method and the number of training samples is studied on the ANN model prediction. The results show that Sobol sequence sampling is superior to other sampling methods especially for low number of samples. The best sampling method is utilized to generate the training dataset with which the hyperparameters of the learning algorithm are optimized. The optimized ANN model is ultimately used to predict the system solar fraction under various design conditions to develop design maps that offer better visualization and sizing guidelines for the hybrid solar thermal storage systems. ANN is shown to offer a potential candidate for accurate and computationally efficient modeling of complex thermal systems. Upon proper configuration and training, the ANN model can accurately (i.e., a coefficient of determination of 0.9999) predict the performance of the hybrid thermal storage system with approximately five orders of magnitude reduction in computational time compared to conventional numerical models.

1. Introduction

Solar radiation is an abundant and clean source of renewable energy; however, its intermittent nature poses limitations on utilizing its full potential. This created significant research attraction toward thermal energy storage as a means to bridge the mismatch between the availability of solar energy and the thermal demand. Water tank storage is the most prominent in domestic heating applications due primarily to design simplicity and low cost. Besides being an abundant and cheap storage medium, water is privileged with superior thermal properties (i.e., heat capacity and thermal conductivity) compared to other storage mediums. The primary drawback manifested by sensible energy storage mediums, like water, when compared to latent energy storage is their lower storage densities, resulting in large operating temperature swings, and thus shorter discharge periods. This necessitates significantly larger tank volumes per unit stored energy.

Phase Change Materials (PCMs) qualify as an energy storage medium

that stores thermal energy in the form of latent heat. This paper focuses on solid–liquid PCMs that melt during the charging periods and solidify during the discharging periods. There is a large body of literature that focuses on the impact of integrating PCMs in solar thermal storage applications. Mazman et al. [1] experimentally investigated the effect of incorporating three different PCM materials at the top of a stratified solar domestic hot water (SDHW) storage tank. It was shown that paraffin wax and stearic acid provided significant enhancement in the thermal performance of the SDHW system. Cabeza et al. [2] incorporated cylindrical granular graphite PCM compound in a stratified SDHW tank while varying the number of PCM modules. Their results showed that increasing the number of PCM modules increased the storage density thus allowing the use of smaller tanks for the same purpose. Mehling et al. [3] showed that integrating cylindrical PCM modules at the top of a storage tank increased the storage density by 20–40%. Abdelsalam et al. [4] showed that introducing PCM modules in solar domestic hot water tanks resulted in around 40% reduction in tank volume compared to sensible storage. Although integrating PCM modules in the storage tanks

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<https://doi.org/10.1016/j.applthermaleng.2021.118009>

Received 4 September 2021; Received in revised form 21 November 2021; Accepted 27 December 2021

Available online 30 December 2021

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Nomenclature*Symbols:*

A_c	Solar collector area [m ²]
C_w	Water specific heat capacity [J/kgK]
F'	Collector efficiency factor
F_R	Collector heat removal factor
$\dot{m}$	Collector mass flow rate [kg/s]
n	Number of data points
R^2	Coefficient of determination
T_l	Load setpoint temperature [K]
T_p	PCM melting temperature [K]
T_w	Water mains temperature [K]
U_L	Collector loss coefficient [W/m ² K]
U_{st}	Tank loss coefficient [W/m ² K]
V_f	PCM volume fraction
V_{st}	Tank volume [L]
X	Raw input data point
$\bar{X}$	Average of input data
X'	Normalized input data point
y	Actual output
y'	Predicted output

ΔT_{off}	Temperature difference at which the collector circulation pump is shut down [K]
ΔT_{on}	Temperature difference at which the collector circulation pump is operated [K]

Greek Symbols:

α	Absorptance
ρ_p	PCM density [kg/m ³]
ρ_w	Water density [kg/m ³]
σ	standard deviation of input data
τ	Transmittance

Acronyms:

ANN	Artificial neural network
LHS	Latin hypercube sampling
LR	Learning rate
MAE	Mean absolute error
MC	Monte Carlo sampling
MSE	Mean squared error
PCM	Phase change material
RMSE	Root mean squared error
SF	Solar fraction

seems appealing, it poses multiple challenges due to the inherently low sensible heat capacity and thermal conductivity of PCMs.

In addition, there is still a gap in the literature about the suitability of PCMs and the optimum design parameters such as PCM volume fraction, melting temperature, and module geometry for various applications, especially solar thermal systems. Suresh and Saini [5] experimentally investigated the effect of varying PCM volume fractions (i.e., 20 – 80%) on the performance of thermal energy storage during charging and discharging periods. The results showed an increase in the charging and discharging periods with the increase of PCM volume fraction. Besides, larger PCM volume fractions were shown to exhibit improved thermal stratification inside the tank. Najafian et al. [6] numerically simulated the performance of a thermal storage tank with PCM modules. Artificial neural network (ANN) was used to correlate the design parameters and the discharge time. The results showed that the best location of PCM modules was near the top. Also, increasing the amount of PCM inside the tank prolonged the discharging periods.

To fully harness the benefits of PCM integration, the effect of the key design parameters of the thermal storage system such as storage tank volume, PCM melting temperature, and PCM volume fraction must be investigated. Detailed numerical modeling and simulation of such systems can be computationally expensive due to the transient dynamic behavior of the storage system and the non-linearity introduced by the phase change process. The power of ANNs can be utilized to provide computationally efficient, yet accurate modeling of solar thermal storage system, thus offering effective means for design optimization. Although ANN existed for a long time, they were implemented for predicting solar systems performance in the late 90's. Alzahrani et al. [7] utilized recurrent neural networks to predict total solar irradiance with a root mean squared error of 0.0513 and 0.0680 for training and testing datasets, respectively. Ozgoren et al. [8] utilized ANN to estimate the monthly mean daily sum global solar radiation in Turkey with a mean absolute percentage error of 5.34% and an R^2 value of 0.9936. Dorvlo et al. [9] utilized ANN methods like radial basis functions (RBF) and multi-layer perceptron (MLP) to predict the total solar irradiance in Oman with both methods showing similar performance. However, RBF method was recommended as it required less computational power and time.

Other researchers attempted utilizing ANN models to predict the

performance of solar thermal energy storage systems. Sridharan and Shenbagaraj [10] utilized generalized regression neural networks (GRNN) to predict the power output and collector efficiency of a standalone, series connected, and parallel connected solar flat plate collector. The results showed a prediction accuracy exceeding 97.64% for collector power output and 94.80% for collector efficiency. Karaağaç et al. [11] experimentally studied a concentrated photovoltaic-thermal solar dryer using nano-enhanced PCM. Moisture content and moisture ratio were predicted using two learning machine algorithms. It was shown that the ANN method outperformed the support vector machines (SVM) method. Motahar [12] experimentally investigated the melting process of a rectangular PCM enclosure subjected to a uniform heat flux boundary condition. A multi-layered perceptron feed-forward neural model was used to predict the Nusselt number and the melted fraction of the PCM. The ANN model predicted the PCM melting characteristics with a mean square error of 4.42×10^{-6} and a correlation coefficient value of 0.999. Kalogirou et al. [13] implemented ANN to predict the daily average energy output as well as the storage tank temperature levels for large solar storage systems. The results showed that the ANN predicted the daily average energy output and tank temperature with R^2 higher than 0.95 and 0.96, respectively. Yaici and Entchev [14] deployed a back-propagation ANN to forecast the performance of a solar thermal energy system for domestic hot water and space heating. The ANN model predicted the stratification temperatures and solar fraction with accuracies of $\pm 3\%$ and $\pm 10\%$, respectively. Géczy-Víg and Farkas [15] utilized a multi-layer perceptron back-propagation neural network to predict the layer temperature of a solar domestic water heating tank using MATLAB neural network toolbox. The ANN model provided predictions with average deviations of 0.22°C and 0.24°C for training and validation datasets, respectively. Kalogirou [16] utilized group method data handling neural networks and genetic algorithm to optimize the collector area and the tank size for a solar industrial process heat system with the objective of maximizing life-cycle savings. The training R^2 was 0.9986.

The literature shows promising utility of ANNs in accurate modeling dynamic coupled thermal systems. Nevertheless, there is still lacking in research studies and investigations on the applicability of ANN modeling to PCM-integrated solar-thermal storage system. This article presents an extensive framework for configuring, training, and testing an ANN

model for a water-PCM solar thermal storage system for domestic water heating applications. A validated numerical model is used to generate the training and testing datasets for the ANN model. The current study demonstrates a deep dive into the configuration and tuning of an open-source ANN model, exploring the effects of different sampling methods, number of samples per dataset, and hyperparameters tuning on the predictive capability of the ANN model. The tuned model is then deployed to create a design map for the solar thermal storage system that illustrates the effects of the tank volume and the PCM volume fraction and melting temperature on the efficiency (i.e., solar fraction) of the solar thermal storage system.

2. System description

The current study uses the validated numerical model developed by Abdelsalam et al. [4,17] for the water-PCM thermal storage system (referred to herein as the hybrid storage system) to generate the training and testing cases for the ANN model. Fig. 1 shows the layout of the hybrid storage system connected to a flat-plate solar collector loop and a demand-side loop. The hybrid storage tank is equipped with an immersed coil heat exchangers at the bottom for charging and another at the top for discharging. The solar collector side is considered the charging loop and the demand side is considered the discharging loop. Heat collected by the solar collector is transferred to the storage tank via the bottom coil heat exchanger. The stored energy is recovered and delivered to the load via the top coil heat exchanger. Water circulates in the collector loop by means of a circulation pump operated by an ON/OFF controller; the pump is operated when there is a sufficient temperature difference between the collector plate and the return from the bottom of the storage tank [18]. The following equation governs the relationship between ΔT_{on} and ΔT_{off} :

$$\Delta T_{off} \leq \frac{A_c \times F_R \times U_L}{\dot{m} \times C_w} \times \Delta T_{on} \quad (1)$$

ΔT_{on} and ΔT_{off} are the temperature difference between the collector plate and the return from the bottom of the storage tank at which the circulation pump is turned ON and OFF. Whereas F_R is the collector heat removal factor calculated using [18]:

$$F_R = \frac{\dot{m} \times C_w (1 - e^{-\frac{A_c \times U_L \times F'}{\dot{m} \times C_w}})}{A_c \times U_L} \quad (2)$$

The area of the collector is denoted as A_c , whereas $\dot{m}$, U_L and F' represent the collector mass flow rate, loss coefficient and efficiency factor, respectively. The heat removal factor F_R is then multiplied by the transmittance-absorptance product ($\tau\alpha$) and Collector loss coefficient

(U_L) to produce $F_R\tau\alpha$ and F_RU_L values respectively. An overheating limiting control is applied on the supply from the collector to prevent the supply temperature to the tank from exceeding 90 °C thus prevents boiling in the collector loop. The load side is fitted with an auxiliary heater and a tempering valve to ensure the temperature of the water delivered to the load does not deviate from the set point. If the delivered water temperature exceeds the setpoint temperature; then the tempering switch will allow mains water to mix with the hot water before being delivered to the load side. However, if the delivered water temperature is lower than the setpoint temperature; then the auxiliary heater provides the required energy to raise the temperature to the required value. Cylindrical PCM modules, 20 mm in diameter, are installed vertically inside the storage tank.

3. System parameters

In this study the system parameters are varied over a wide range of values to investigate the ANN predictive capability. The area of the solar collector ranges from 1 to 8 m². The storage tank volume is varied from 50 to 240 L, and the setpoint temperature for the hot water delivered to load is varied from 55 to 60 °C. Table 1 summarizes the proposed system parameters for the current study.

A python script is developed to simulate multiple cases with various operating conditions. The following parameters are varied: collector area, tank volume, load temperature, PCM melting temperature, and PCM volume fraction (the ratio of PCM volume to the total tank volume). The solar radiation and ambient temperature profiles are obtained from (https://weather.gc.ca/canada_e.html) for a typical spring day in the city of Toronto, Canada, shown in Fig. 2. A dispersed hot water demand profile [19] is assumed to represent a typical household usage with a fixed draw rate of 8 L/min and a daily total of 189 L, shown in Fig. 3.

4. Artificial neural network modeling

Artificial neural networks are data processing systems capable of analyzing and correlating input and output data. The fundamental elements that make up the ANN are called neurons. These neurons are interconnected with synaptic weights. Each neuron applies certain mathematical functions to its inputs and then communicates the results with other neurons as shown in Fig. 4. ANN has the capability of recognizing patterns of multi-dimensional complex datasets and providing a generalized correlation between the inputs and outputs that enables predicting the output of new data points. For more details about ANN modeling, refer to [20].

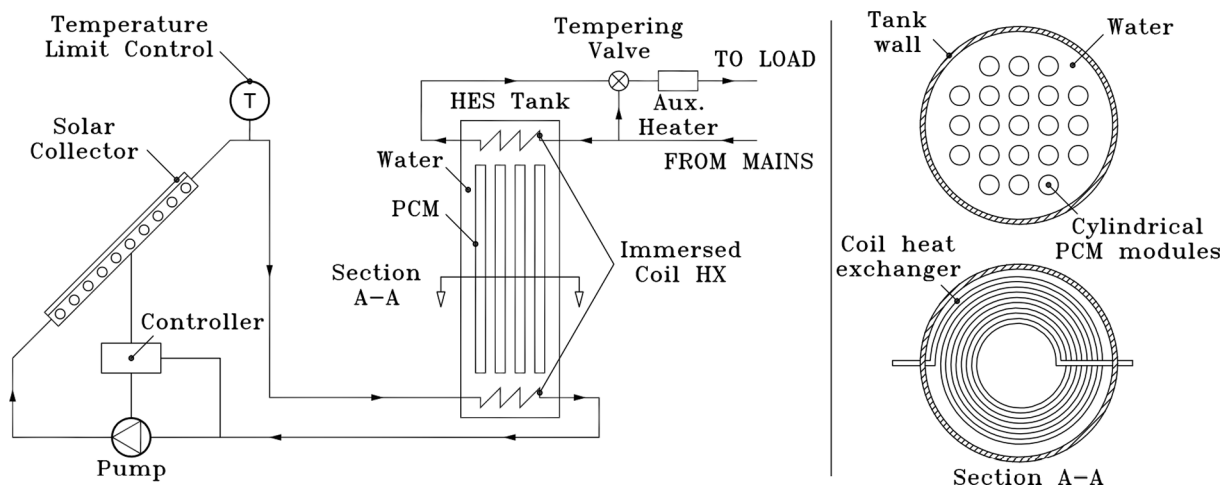


Fig. 1. Solar hybrid thermal storage system with cylindrical PCM modules and immersed coil heat exchangers. Adapted from Abdelsalam et al. [4].

Table 1
Proposed system parameters.

Parameters	Value
Solar collector	
Collector area (A_c)	1 to 8 [m ²] ^a
Collector mass flow rate ($\dot{m}$)	0.05 [kg/s]
Collector fluid specific heat capacity (C_w)	4.18 [kJ/kgK]
ΔT_{off}	2 [K] ^b
Transmittance-absorptance product ($\tau\alpha$)	0.8 ^c
Collector loss coefficient (U_L)	5.0 [W/m ² K] ^c
Collector efficiency factor (F')	0.84 ^c
Storage tank	
Storage tank volume (V_{st})	50 to 240 [L]
Tank aspect ratio	2:1
Number of tanks	1
Tank surrounding temperature (T_a)	20 [°C]
Tank loss coefficient (U_{st})	0.5 [W/m ² K]
Phase change material (PCM)	
PCM Latent heat	182 [kJ/kg] ^d
PCM Density (ρ_p)	870 [kg/m ³] ^d
PCM Melting temperature (T_p)	25 to 35 [°C] ^d
PCM Volume fraction (V_f)	0.0 to 0.7
Water properties	
Load temperature (T_l)	55 to 60 [°C]
Water Mains temperature (T_w)	8 [°C]
Water density (ρ_w)	993 [kg/m ³]
Water conductivity	0.63 [W/mK]

^a Assuming 0.35–2.67 m²/person.

^b This value was used to minimize the cycling frequency on the circulation pump [18].

^c The values were calculated based on 45° slope flat plate collector facing south [18].

^d The PCM properties are similar to Capric acid [33].

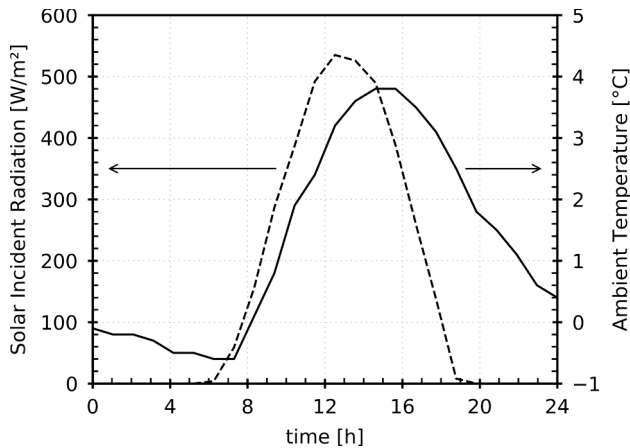


Fig. 2. Hourly solar incident radiation and dry bulb temperature for a typical spring day in the city of Toronto, Canada.

a. Data pre-processing

Raw data often requires pre-processing and transformation before introduction to neural networks. Scaling and normalization of the data are often required to ensure the stability and robustness of the ANN training [21]. The following formula is used to normalize and scale the training data:

$$X' = \frac{X - \bar{X}}{\sigma} \quad (3)$$

where, X' is the normalized input data point, X is the raw input data point, $\bar{X}$ is the raw dataset average, and σ is the raw input dataset

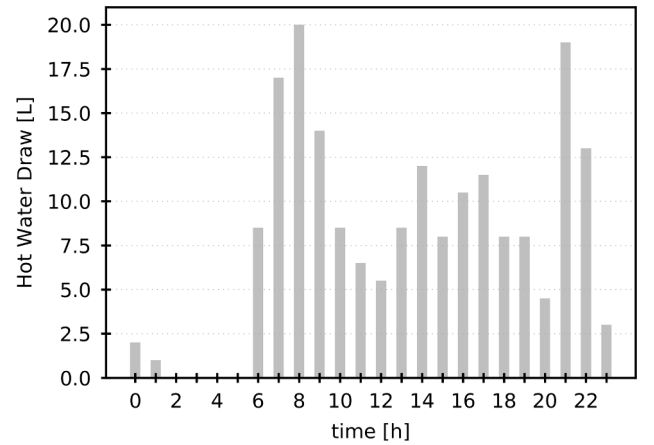


Fig. 3. Dispersed hot water draw profile reported by [19], scaled to a total daily demand of 189 L. The time “0” represents 12:00 AM.

standard deviation. The processed data is then sampled to create a training set for the ANN model. The literature presents various data sampling techniques for the training sets. This study explores using three different sampling methods: the Monte Carlo (MC), the Latin hypercube (LHS), and the Sobol sequence sampling.

b. Training process

After data pre-processing, the ANN model is introduced to the entire training dataset and the synaptic weights are randomly initialized to predict an output value for each data point. The output is then subjected to a predefined loss function, and the error between the actual and the predicted values is calculated. Among the common loss functions are the mean squared error (MSE), the root mean squared error (RMSE), and the mean absolute error (MAE). Eqs. (4–6) present the mathematical definitions of the three loss functions. The optimizer (also referred to as the learning algorithm) adjusts the synaptic weights with the purpose of minimizing the loss function, this is considered as one “epoch”.

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n} \quad (4)$$

$$MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n} \quad (5)$$

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (6)$$

where, n is the number of data points and y_i and $\hat{y}_i$ are the actual and the predicted output values, respectively. The optimizer performs a predefined number of epochs with the target of improving the accuracy of the model. However, excessive increase of the number of epochs often leads to model overfitting, by which the increase in the validation loss outweighs the decrease in the training loss. That is, the overfitted ANN model performs extremely well on training data but underperforms on new data, thus loses its generality [22].

c. Development of ANN models

KERAS and TensorFlow, open-source ANN modeling packages developed by Google, are used in this study [23]. A feed-forward multi-layer ANN type is chosen for this study [24]. A particular difficulty that is sometimes overlooked in current machine learning applications is the optimization of internal model parameters, also referred to as the hyperparameters such as the learning rate, the number of hidden layers,

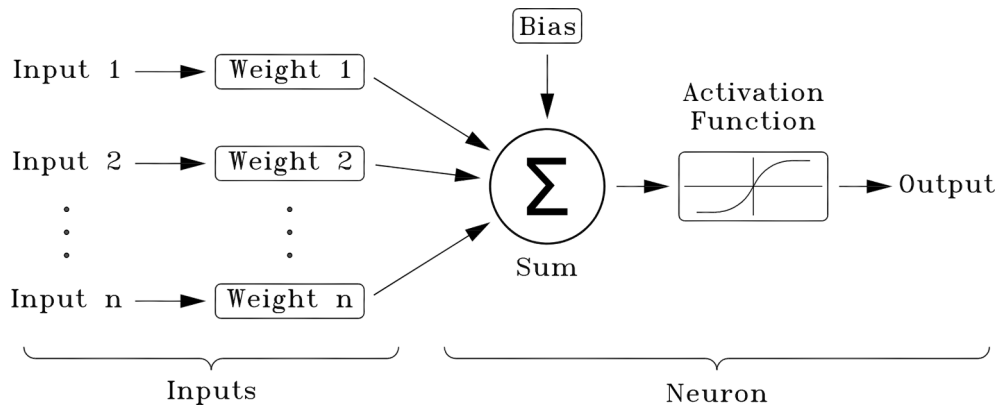


Fig. 4. ANN diagram showing the relationship between neuron inputs and output. The neuron output is the sum of all inputs multiplied by synaptic weights.

the number of neurons per hidden layer, and the activation functions [25]. Proper configuration of the model hyperparameters is key to the performance and the predictive capabilities of the ANN model. As such, the current study investigates the impact of hyperparameters tuning on the prediction accuracy of the thermal storage system performance. The metric of performance of the solar thermal storage system is chosen to be the solar fraction (SF) defined as the ratio of the thermal energy delivered to the load to the total thermal demand:

$$\text{SolarFraction(SF)} = \frac{\text{Thermal energy delivered to the load}}{\text{Total thermal demand}} \quad (7)$$

The solar fraction (SF) for solar thermal storage system approaches unity as higher fraction of the thermal demand is satisfied by the solar thermal storage system rather than the auxiliary heater.

5. Results and discussion

This section is organized to show the effect of different sampling techniques, number of samples, and hyperparameters tuning on the predictive capabilities of the neural network. Then, the utility of the optimized ANN model is demonstrated through the generation of design maps for the solar hybrid storage system.

a. Sampling method and number of samples

For the ANN model to accurately simulate the solar thermal storage system, it must train on a set of data strategically selected to cover the studied range summarized in Table 1. Each data point (training sample) contains different values for the five input parameters (collector area, storage tank volume, load temperature, PCM melting temperature, and PCM volume fraction) and the corresponding output (solar fraction). As such, there are two parameters that need to be carefully determined: which sampling method to be utilized to sample the studied range and what is the minimum number of training samples required to adequately train the ANN model. Three sampling techniques were investigated: Sobol sequence, Latin hypercube, and Monte Carlo. A python code is developed to sample the studied range. A training set up to 10,000 training samples was generated utilizing each sampling method. ANN model was created and trained using each sampling method at the following number of training sample increments (250, 500, 1000, 1500, 2000, 3000, 4000, 5000, 6000, 7000, 8000, 9000, 10000) totaling 13 models for each sampling method (39 models overall). A comprehensive testing set of 84,480 samples covering the entire studied range was created (these testing samples were not introduced to the ANN models during the training phase and thus will test the predictive and generalization ability of each model). The testing MAE, RMSE, and the coefficient of determination (R^2) for the sampling methods are plotted against the number of samples, as shown in Figs. 5–7, respectively.

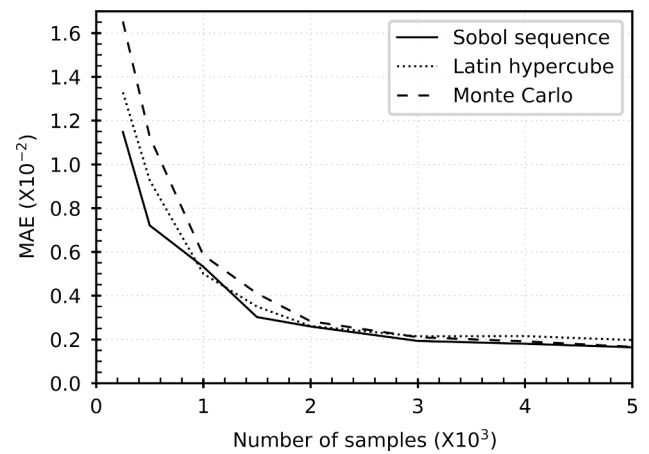


Fig. 5. Mean absolute error (MAE) for the three sampling methods versus the number of samples.

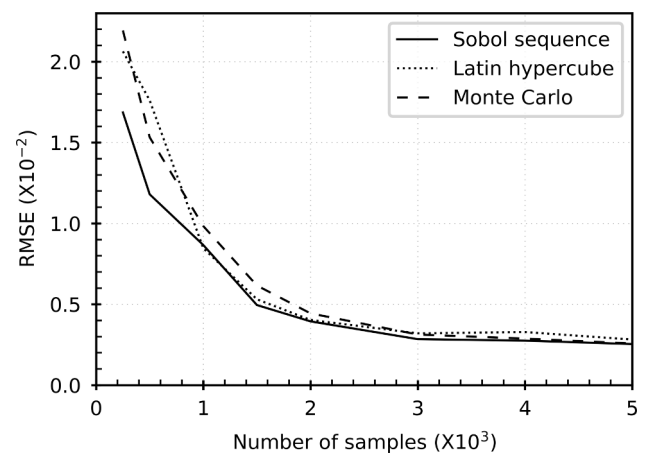


Fig. 6. Root mean squared error (RMSE) for the three sampling methods versus the number of samples.

It is observed that the Sobol sequence sampling outperforms other sampling techniques, providing 44% and 16% reduction in the testing mean absolute error (MAE) compared to LHS models and MC models, respectively. This is found to agree with Burhenne et al. [26] whose results showed favorable performance of the Sobol-based sampling over other techniques. Increasing the number of training samples results in a more accurate ANN predictions. However, In the limit of larger number of samples, the different sampling techniques provide similar prediction

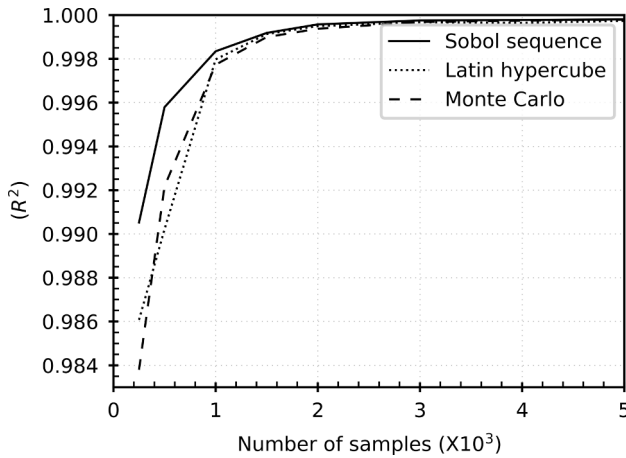


Fig. 7. Coefficient of determination (R^2) for the three sampling methods versus the number of samples.

accuracy among the generated ANN models. It is observed that increasing the number of training samples beyond 3,000 samples slightly reduces the MAE.

b. Hyperparameter optimization

Hyperparameter optimization is the process of configuring the ANN hyperparameters. An ANN model with a properly configured hyperparameters will lead to an accurate ANN predictions (the ANN will mimic the studied system accurately). The optimization of the model hyperparameters is carried out for the Sobol sequence sampling ANN model with a total of 3,000 training samples. The optimizer is a machine learning algorithm used to adjust the synaptic weights during the training process with the goal of minimizing the loss function. Adaptive moment estimation learning algorithm, known as “Adam” algorithm [27], is simple and computationally efficient for gradient-based optimization of stochastic objective functions. It is well poised for machine learning problems with large datasets and/or multi-dimensional parameter spaces. As such, this study employs the Adam learning algorithm to minimize the loss function. A methodical approach is developed to optimize the key hyperparameters of the learning algorithm: the learning rate, the number of hidden layers, the number of neurons per hidden layer, and the activation function, to enhance the predictive capability of the ANN model.

i. Learning rate

The learning rate (LR) in machine learning is the step size after each epoch towards minimizing the loss function. Higher learning rates often results in faster training on the expense of model training stability. Conversely, smaller learning rates yields slower yet more stable training process, but it might cause the loss function to get stuck in an undesirable local minimum [28]. Therefore, an optimum learning rate is required to reach the global minimum value of the loss function while preserving the model training stability. Table 2 shows the LR optimization process. A learning rate of 0.005 is seen to achieve the lowest

Table 2
ANN testing performance for different learning rates.

Learning rate	Testing performance		
	MAE	RMSE	R^2
0.100	0.00248	0.003474	0.99960
0.010	0.00203	0.003099	0.99973
0.005	0.00190	0.002729	0.99976
0.001	0.00395	0.006725	0.99897

testing error.

ii. ANN topology

ANN consists of layers of interconnected neurons. A typical ANN consists of an input layer, one or multiple hidden layers, and an output layer. Varying the number of hidden layers and the neurons in each hidden layer has a great effect on the performance of ANN models [29]. Lower number of hidden layers simplifies the model on the expense of its ability to effectively capture the complexities and non-linearities in the modelled system. On the other hand, increasing the number of hidden layers beyond a certain value can lead to model overfitting. Table 3 shows the effect of varying the number of hidden layers from 1 to 4 and the number of neurons in each hidden layer to 30, 40, and 50. It is observed that 3 hidden layers with 50 neurons in each hidden layer provide the optimum structure to achieve improved predictions.

iii. Activation function

Activation functions also known as “transfer functions” are used to limit the neuron output within a reasonable range of values [30]. Without activation functions the ANN will simply be a linear regression model. Activation functions are applied after the mathematical processes in the neuron. There is a variety of activation functions available in the literature, each having its advantages and limitations. For example, rectified linear activation function (RELU) is computationally efficient but not suitable for inputs that approach zero or negative values, whereas Softmax activation function is suitable for classification problems. Table 4 shows a comparison of the ANN model performance under various activation functions. It is observed that sigmoid activation function is the most suitable for this study.

iv. Single vs. Multi-ANN models for prediction

To further improve the predictive capabilities of the ANN model, the best three models during the training and the testing stages are selected for prediction. The outlier prediction out of the three is eliminated for each data point and the average of the other two is considered as the output. This method further reduces MAE by approximately 10% and the maximum absolute error by 21% compared to single ANN model predictions, shown in Table 5. This approach reduces the maximum error significantly while the improvements in MAE are minor. This is possible since the maximum error for each ANN model occurs at different input values. Fig. 8 shows a good agreement between the actual and the predicted values of the solar fraction of the storage system as all values are near the line of equal actual and predicted solar fraction.

Table 3
ANN testing performance for different ANN topologies.

Hidden layers	Neurons	Testing performance		
		MAE	RMSE	R^2
1	30	0.00388	0.006226	0.99912
	40	0.00356	0.005735	0.99922
	50	0.00387	0.006133	0.99909
2	30	0.00183	0.002662	0.99978
	40	0.00208	0.002994	0.99976
	50	0.00173	0.002498	0.99980
3	30	0.00175	0.002476	0.99979
	40	0.00145	0.002081	0.99985
	50	0.00142	0.002044	0.99986
4	30	0.00169	0.002443	0.99981
	40	0.00159	0.002205	0.99985
	50	0.00142	0.002103	0.99985

Table 4

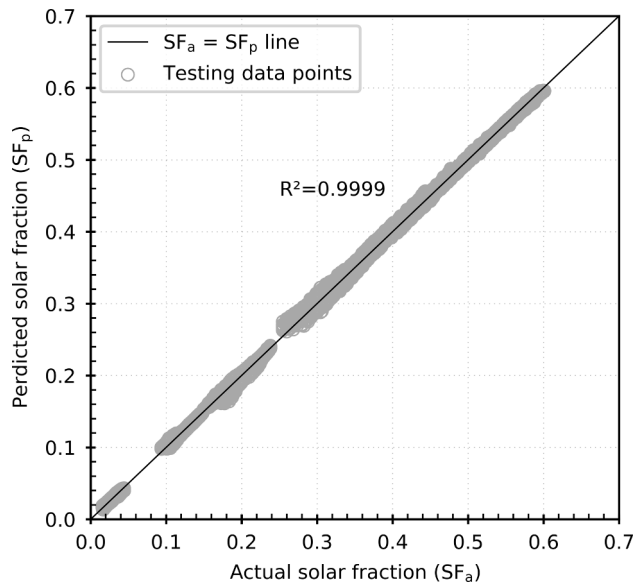
ANN testing performance for different activation functions.

Activation function	Formula		Testing performance		
			MAE	RMSE	R ²
RELU [34]	$f(x) = \begin{cases} 0, & x \leq 0 \\ x, & x > 0 \end{cases}$	(8)	0.00165	0.002717	0.99976
sigmoid	$f(x) = \frac{1}{1 + e^{-x}}$	(9)	0.00126	0.001943	0.99987
Softplus [35]	$f(x) = \ln(1 + e^x)$	(10)	0.00155	0.002683	0.99977
tanh	$f(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$	(11)	0.00167	0.002565	0.99979
SELU [36]	$f(x) =$	(12)	0.00238	0.003294	0.99967
ELU [37]	$\begin{cases} \alpha(e^x - 1), & x \leq 0 \\ x, & x > 0 \end{cases}$	(13)	0.00159	0.002369	0.99982
	$\lambda \begin{cases} \alpha(e^x - 1), & x \leq 0 \\ x, & x > 0 \end{cases}$				

Table 5

ANN testing performance for single- versus multi-ANN model.

ANN model	Testing performance			
	MAE	RMSE	MAX. MAE	R ²
Single ANN	0.00126	0.001943	0.0256	0.99987
Multi-ANN	0.00114	0.001745	0.0203	0.99990

**Fig. 8.** Actual versus predicted solar fraction.

v. Data fitting and correlations

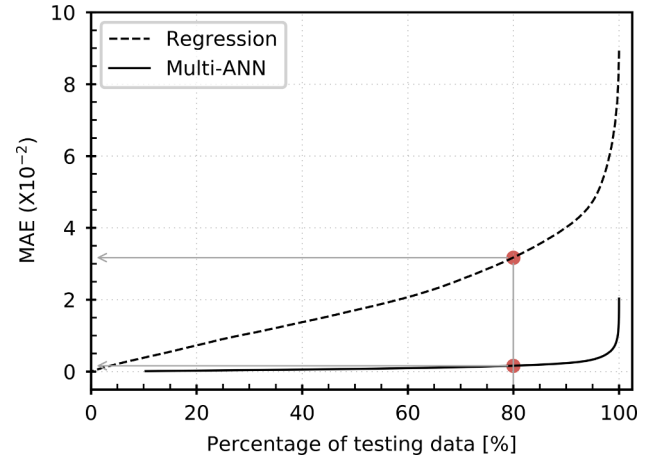
A regression model is developed to fit the training data and provide a correlation between the system solar fraction and the controllable variables: the collector area, the storage tank volume, the load temperature, the PCM melting temperature, and the PCM volume fraction. The following equation represents the regression model:

$$\begin{aligned}
 SF = & -0.009248A_c^2 + 0.000055A_cV_{st} - 0.001532A_cT_l \\
 & + 0.000589A_cT_p + 0.001477A_cV_f - 0.000003V_{st}^2 + 0.000004V_{st}T_p \\
 & - 0.000242V_{st}V_f - 0.000068T_p^2 + 0.002308T_pV_f - 0.04611V_f^2 \\
 & + 0.21993A_c + 0.000779V_{st} + 0.000112T_l + 0.00086T_p - 0.01V_f - 0.1908
 \end{aligned}
 \quad (14)$$

Table 6

Testing performance for multi-ANN versus regression model.

Model type	Testing performance			
	MAE	RMSE	MAX. MAE	R ²
Multi-ANN	0.00114	0.001745	0.0203	0.99990
Regression	0.02010	0.024866	0.0893	0.98481

**Fig. 9.** Mean absolute error distribution for Multi-ANN versus regression model.

where, A_c = collector area, V_{st} = storage tank volume, T_l = load temperature, T_p = PCM melting temperature, and V_f = PCM volume fraction. The results are summarized in Table 6 and Fig. 9 represents the MAE distribution for ANN versus regression model which highlights the superiority of ANN modeling over regression. Fig. 9 shows that 80% of the predicted testing data has a MAE less than 2×10^{-3} for the ANN model, whereas 80% of the predicted testing data for the regression model has a MAE less than 3.2×10^{-2} .

c. Design maps for the solar hybrid storage system

The ANN model is deployed to predict the solar fraction (SF) for the hybrid storage system with different tank volumes, PCM volume fractions, and PCM melting temperatures. Table 7 presents the optimized model hyperparameters used in this study. Fig. 10 shows the selected ANN topology.

Mapping the solar fraction contours with respect to the various design parameters of the hybrid storage system requires numerous simulations that are computationally intensive. Such a challenge can be circumvented by the utilization of a well-trained ANN model that is capable of generating high-resolution performance contours of the solar storage system with high accuracy and significantly lower computational effort. Fig. 11 presents the solar fraction contours for different storage tank volumes and PCM volume fraction where the collector area

Table 7

Optimized ANN model hyperparameters.

Hyperparameter	Selected value
ANN type	Feed-forward multi-layer
Number of input neurons	5
Number of output neurons	1
Number of hidden layers	3
Neurons per hidden layer	50
Learning algorithm (optimizer)	Adam
Learning rate	0.005
Activation function	sigmoid

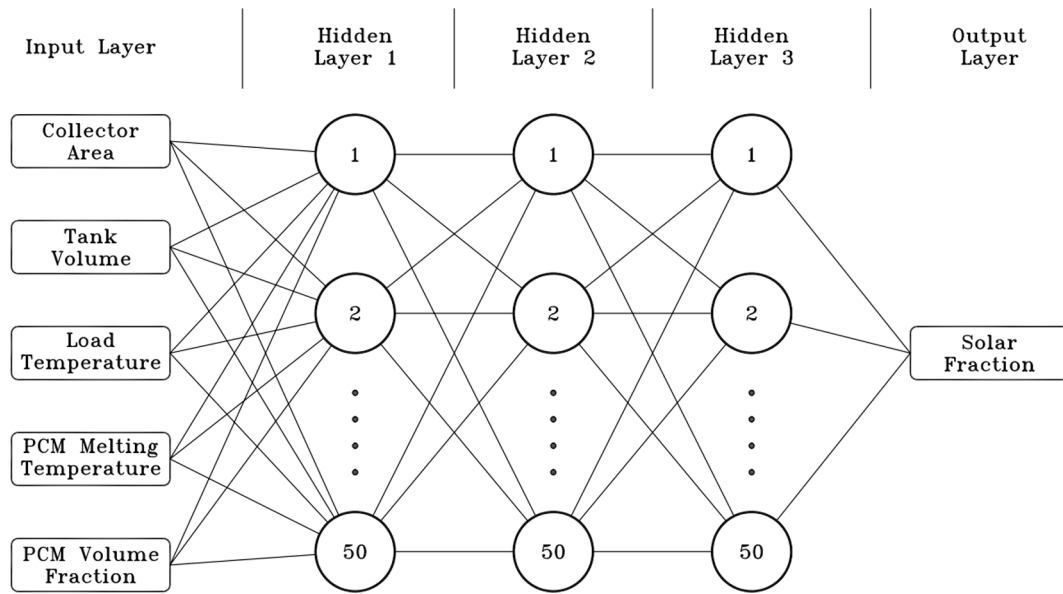


Fig. 10. ANN topology showing input, hidden, and output layers and neurons.

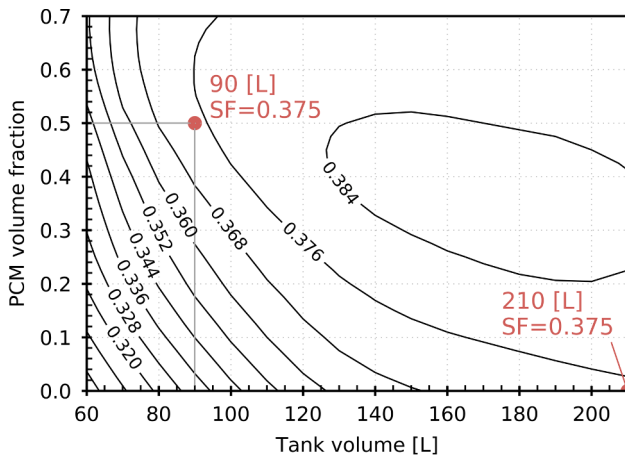


Fig. 11. Solar fraction contours for various PCM volume fractions and tank volumes. The collector area is 4 m^2 , the load temperature is 55°C and the PCM melting temperature is 28°C .

is fixed at 4 m^2 , the load temperature is 55°C and the PCM melting temperature is chosen at 28°C . Following a horizontal line on the contour plot shows the solar fraction approaching a peak with the increase of the tank volume followed by a drop. Such a behavior was proved in previous studies to be due to effect of the increased thermal losses associated with the larger surface area of the larger storage tanks [31].

Fig. 12 shows the solar fraction contours for different PCM melting temperature and PCM volume fraction while keeping collector area and load temperature constant at 4 m^2 and 55°C , respectively. It is shown that for a chosen PCM melting temperature of 28°C , increasing the PCM volume fraction from 0 (Point A on the left plot of Fig. 12) to 0.5 (Point B on the left plot of Fig. 12) in a 90-L tank increases the solar fraction by about 13%, compared to around 5% for a 150-L tank (Points C and D on the right plot of Fig. 12). This highlights that integrating PCM modules in the storage tank increases the storage density of the tank, resulting into prolonged discharging periods, hence improving the solar fraction. However, such an improvement is more prominent in lower tank volumes due primarily to the adverse effects of thermal losses and larger storage capacity associated with larger tank volumes [32]. The contour plots in Fig. 12 also show an optimum value of PCM volume fraction beyond which the system solar fraction drops. This is attributed to the inherently low thermal characteristics (i.e., thermal conductivity and specific heat capacity) of PCM compared to water, often resulting into

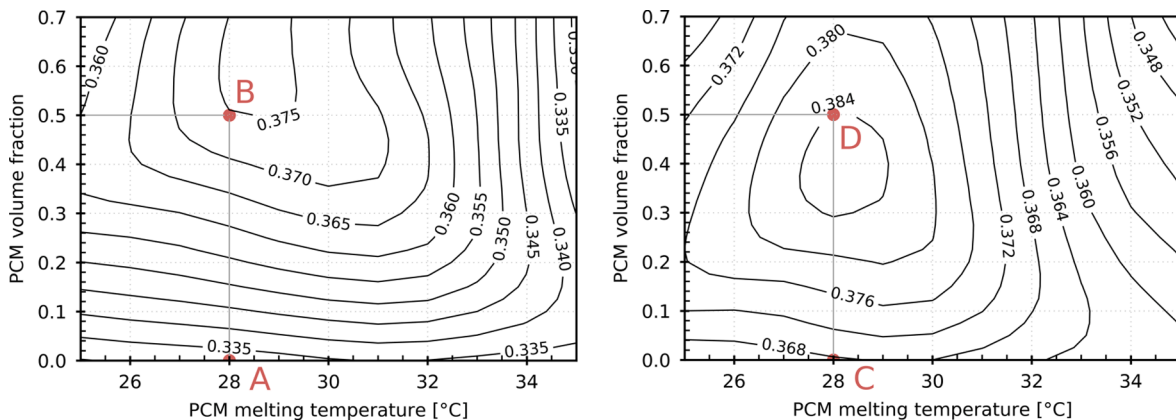


Fig. 12. Solar fraction contours for various PCM volume fractions and melting temperatures. The collector area is 4 m^2 and the load temperature is 55°C . Left: the tank volume is 90 L, Right: the tank volume is 150 L.

incomplete melting and solidification of the PCM during the charging and discharging cycles. This is found in agreement with the results by Suresh and Saini [5]. Improper selection of the PCM melting temperature also contributes to partial phase transformation during the charging and discharging processes, thus hindering the full utilization of the PCM latent heat capacity. As such, the majority of the PCM will act as a sensible heat storage medium with inherently inferior thermal characteristics compared to water, thus degrading the overall system performance. This shows that the benefits of incorporating PCM modules in storage tank is noticeable at low storage tank volumes. In fact, if PCM melting temperature and volume fraction are selected properly, PCM integration can yield up to 57% reduction in tank volume. As seen in Fig. 11 incorporating PCM modules in a 90 L tank yields the same solar fraction as a 210 L tank without PCM.

6. Conclusions

The potential of ANN modeling is extensively investigated for a hybrid solar thermal storage system involving PCMs for domestic water heating applications. The effects of using different sampling methods, number of samples, and hyperparameters tuning are studied. A total of 39 ANN models are developed to investigate the effect of sampling methods and number of samples on the predictive capabilities of the ANN models. The results show that the Sobol sequence sampling provides less overall error in the testing sets compared to the Latin hypercube and the Monte Carlo sampling methods, especially for low number of data points in the training set. Hyperparameter optimization shows the effect of tuning the ANN hyperparameters on the predictive capabilities of the ANN models. The results show that ANN can be used successfully to model the hybrid solar thermal storage systems with an R^2 value of 0.9999. The optimized ANN model is then deployed to generate design maps for the system, offering accessible sizing and selection guidelines for the key design parameters: the storage tank volume, the PCM volume fraction and the PCM melting temperature. PCM parameter visualization shows that proper selection of the tank volume and the PCM melting temperature and volume fraction is essential to improve the system solar fraction. One of the main benefits of using ANN modeling is the significant reduction in the computational effort. The time spent to simulate the entire testing set (84,480 testing samples) using the numerical model by Abdelsalam et al. [4] is recorded to be over 120 h compared to around 5 s using the ANN model. As such, with careful configuration and training, ANN modeling can provide computationally efficient means to predict the performance of highly transient and non-linear systems. The framework presented in this article can be extended to involve other design and operational parameters such as the solar collector area, the weather profile, the demand profile to develop more comprehensive design maps for solar thermal storage systems.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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